

Seat Majorities without Vote Majorities: Coalition Inversions in Brazil’s Chamber of Deputies

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Abstract

Proportional representation is usually evaluated at the party level, asking whether seats track votes. In fragmented legislatures, however, the politically operative unit is not a party but a coalition, and the aggregation of party-level seat differentials can flip the majority status of a party set without flipping its vote share. This article asks whether Brazilian open-list proportional representation produces coalition-level election inversions; instances in which a set of parties holds a seat majority in the Chamber of Deputies while corresponding to less than half of the national federal-deputy vote. Using party vote totals, elected-seat totals, cabinet-period composition, and ideology classifications for the 2014, 2018, and 2022 elections, I identify inversions in two analytically distinct objects: observed cabinet coalitions and ideologically contiguous party intervals. Three cabinet periods satisfy the inversion criterion. Two arise from the 2014 election; a two-day transition window during the impeachment rupture and the substantively important Temer-period coalition, which held 266 seats with 49.91 percent of the vote. One arises from the 2022 election: the initial Lula coalition held 263 seats with 48.89 percent of the vote. No observed inversion appears after the 2018 election, because Bolsonaro-period coalitions do not cross the 257-seat threshold. The ideological interval search confirms the pattern and reveals a structurally striking case in 2022: a compact right-side interval of eight parties from PP to PL holds 258 seats with only 45.35 percent of the vote. The results establish that Brazilian proportional representation can translate nationally minority party sets into legislative majorities. The finding is an institutional claim about the mechanistic properties of the seat allocation, not an inference about voter preferences over cabinets or the legitimacy of any government.

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1 Introduction

Electoral institutions translate social support into political authority, and this translation is never neutral in a mechanical sense. Votes are counted through a procedure, seats are allocated under rules, and governments are formed from the legislative distribution that results. In proportional representation systems, the central justificatory claim is that the conversion of votes into seats should preserve the distribution of political support more faithfully than plurality or majoritarian systems. A segment of the comparative politics literature has refined this standard by identifying the conditions under which proportional systems nevertheless generate systematic distortions; through district magnitude effects, legal thresholds, and pure rounding error (Mainwaring 1999; Ames 2001; Figueiredo and Limongi 2000; Raile et al. 2011). What this literature has not fully examined, however, is the coalition-level implication of these distortions.

This raises a precise analytical question. Can a coalition of parties command a majority of seats in the Chamber of Deputies while corresponding to less than a majority of the national federal-deputy vote? If so, Brazilian proportional representation does not merely redistribute seat shares relative to vote shares at the party level. It can produce coalition-level inversions, in which the majority relation among party sets in the legislature differs from the majority relation among those same sets in the national vote. The issue is not whether PR fails to give every party its exact seat quota; no finite legislature can do that; but whether the accumulated deviations cross the majority threshold at the coalition level. The distinction matters because it is at the coalition level that parliamentary authority is actually organized.

The social choice literature on PR inversions provides the relevant analytical framework. Inversions at the coalition level can occur even when seat allocations are individually monotonic with respect to party vote shares, because monotonicity with respect to individual parties does not imply monotonicity with respect to all coalitions. A coalition is a post-allocation aggregation of party-level seat outcomes, not a unit that receives seats directly. As Kurrild-Klitgaard shows in the Danish context, PR can generate rank-order discrepancies between vote and seat distributions at the level of coalitions rather than parties (Kurrild-Klitgaard 2008; Kurrild-Klitgaard 2013). Miller sharpens this point formally: under PR, coalitionwise inversions are more likely than partywise inversions precisely because single parties rarely win majorities of both votes and seats in fragmented systems (Miller 2015).

The Brazilian case is useful for operationalizing this problem because the institutional chain from votes to coalition authority contains several distinct aggregation steps, each of which can contribute to the inversion. Federal-deputy seats are allocated in state-level districts, not through a single national district. The party system is highly fragmented, producing a large number of small delegations whose seat differentials can accumulate. Cabinet coalitions are assembled after elections under coalitional presidentialism, through bargaining over portfolios and legislative support. The Chamber majority that sustains or constrains a government is therefore the product of successive aggregations; from votes to state-level party seats, from state-level seats to national

party delegations, and from national delegations to governing coalitions. Evaluating whether any of these coalitions constitutes a seat majority without a vote majority requires holding the aggregation procedure visible throughout.

The analysis proceeds in two stages, which answer different but complementary questions. The first examines observed cabinet-period coalitions. These are the primary empirical objects because they are realized party sets; coalitions that actually organized parliamentary authority during specific periods of government formation or support. For each cabinet period in the mandates following the 2014, 2018, and 2022 elections, I translate cabinet party labels into the election-year party space and compute the coalition’s national vote share, seat share, seat quota, and seat differential. The second stage examines ideologically contiguous intervals. Given a party ordering by ideology, I enumerate every contiguous no-gap interval of parties and identify those that constitute a seat majority without a vote majority. This step does not replace the cabinet analysis; it asks whether the inversion is also embedded in the ideological geometry of the Chamber, independently of which coalitions were realized.

The results are as follows. Three observed cabinet periods satisfy the inversion criterion. The first, a two-day period in April 2016 (2016.2), is best understood as an event-window diagnostic around the impeachment rupture rather than a stable governing configuration. The second, the Temer-period coalition (2017.1), is the substantively important case from the 2014 election: it held 266 seats with 49.91 percent of the national federal-deputy vote and persisted for 100 days. The third is the initial Lula coalition after the 2022 election (2023.1), which held 263 seats with 48.89 percent of the vote across 255 days. Conversely, no Bolsonaro-period coalition after the 2018 election reaches both thresholds, despite some cases of overrepresentation, because none crosses the 257-seat floor. The ideological interval results corroborate the pattern while adding structural content. In both 2014 and 2022 there exist endpoint-minimal contiguous intervals that are inversions; in 2018 there are none, despite 172 seat-majority intervals. The most striking structural result is the 2022 PP–PL interval: eight parties, 45.35 percent of the vote, 258 seats.

The scope of the analysis should be stated clearly before proceeding. The analysis holds background political conditions fixed and isolates the mechanistic properties of the seat allocation procedure. It does not model strategic adaptation under alternative allocation rules, does not estimate voter preferences over coalitions as collective objects, and does not evaluate the legitimacy or stability of any specific government. The claim is institutional and prior to those questions: under the observed vote-seat translation, some party sets become legislative majorities without being national vote majorities.

2 Coalition inversions and proportional representation

A key analytical point to anchor this discussion is that proportional formulas are normally monotonic at the party level but not necessarily at the coalition level. If party *A* has more votes than party *B*, a proportional allocation rule will not ordinarily assign fewer seats to *A* than to *B*, setting aside thresholds and district effects. This monotonicity property, however, applies to individual parties as units of allocation, not to coalitions. Since a coalition’s seat total is the sum of the seat totals of its member parties, and each party’s seat total can differ from its proportional quota in either direction, a coalition can accumulate enough positive differentials to cross the seat-majority threshold without crossing the vote-majority threshold.

The literature has identified this possibility at different levels of generality. Kurrild-Klitgaard’s analysis of Danish elections shows that PR can generate rank-order discrepancies between seat allocations and majority comparisons over parties, and that these discrepancies become more con-

sequential when evaluated at the level of government formations rather than individual parties (Kurrild-Klitgaard 2008; Kurrild-Klitgaard 2013). Miller’s formal treatment clarifies why: in fragmented systems, where single parties rarely command absolute majorities on either dimension, the whole-number problem in seat allocation creates systematic scope for coalitionwise inversions even when partywise inversions are absent or negligible (Miller 2015). The present analysis contributes to this literature by moving from possibility to empirical observation, using realized cabinet coalitions and structured ideological intervals as the relevant objects.

The game-theoretic notion of a coalition as a set of actors (Shapley 1962; Felsenthal and Machover 1998) provides the general framing, but the present analysis distinguishes two senses of coalition that call for separate treatment. Observed cabinet coalitions are coalitions in the narrower political sense: they are party sets that participated in, supported, or were compatible with a specific governing arrangement, and they are identified through historical cabinet composition data rather than constructed post hoc. Ideological intervals are structured coalitions in the set-theoretic sense, constrained by a no-gap ideological ordering. These two objects answer different questions. Cabinet coalitions test whether politically realized party sets were seat majorities without vote majorities. Ideological intervals test whether the vote-seat geometry generates this property for structured regions of the party space, regardless of which coalitions were actually formed.

The Brazilian institutional context amplifies the conditions under which coalition-level inversions can emerge. Open-list PR in state-level districts of varying magnitude produces seat allocations that diverge systematically from national vote shares, with smaller states overrepresented and larger states producing different party-seat ratios than their vote totals would imply under a single national district. High party fragmentation means that governing coalitions must assemble large numbers of small delegations, each with its own seat differential. Coalitional presidentialism creates incentives for presidents to construct oversized legislative coalitions, further expanding the party-set universe over which inversions can be tested (Mainwaring 1999; Ames 2001; Figueiredo and Limongi 2000; Raile et al. 2011). These are background conditions, not the mechanism under analysis. The analysis does not decompose inversions into their district, threshold, and rounding components. It establishes that inversions occur and characterizes their magnitude, duration, and ideological structure.

3 Measurement and data

Let P be the set of parties in a federal-deputy election. For each party $i \in P$, let v_i denote the party’s national federal-deputy vote total, s_i the number of Chamber seats won by that party, and $V = \sum_{i \in P} v_i$ the total national vote. Since the Chamber has 513 seats, the proportional seat quota of party i is

$$q_i = 513 \frac{v_i}{V},$$

and the seat differential is

$$d_i = s_i - q_i.$$

Positive differentials mean the party received more seats than its national vote share would imply under exact proportionality; negative differentials mean it received fewer. Party-level seat differentials sum to zero by construction, since both seats and quotas sum to 513.

For any coalition $C \subseteq P$, define

$$v_C = \sum_{i \in C} v_i, \quad s_C = \sum_{i \in C} s_i,$$

$$q_C = 513 \frac{v_C}{V}, \quad d_C = s_C - q_C.$$

A coalition has a vote majority when $v_C/V > .5$ and a seat majority when $s_C \geq 257$.

Definition 1 (Coalition inversion). *A coalition inversion occurs when a party set C has a seat majority but not a vote majority:*

$$s_C \geq 257 \quad \text{and} \quad \frac{v_C}{V} \leq .5.$$

Equivalently, C commands a Chamber majority without commanding a national federal-deputy vote majority in the election that produced the Chamber.

The definition isolates the majority threshold as the relevant criterion. A coalition can be overrepresented; have a positive seat differential; without constituting an inversion. Overrepresentation becomes an inversion only when the accumulated positive differential crosses the seat-majority floor. This is the analytically important distinction: the criterion separates ordinary proportionality error from a change in the majority status of a party set.

The analysis uses five data objects. First, party vote totals for federal-deputy elections in 2014, 2018, and 2022, drawn from Tribunal Superior Eleitoral filings. Second, elected-seat totals for the same elections. Third, cabinet-period party composition data. Fourth, a cabinet-to-election party crosswalk that translates cabinet labels into the election-year party space. Fifth, party ideology classifications to order parties for the interval analysis.

The vote-column choice requires a substantive decision. The 2014 TSE file schema uses QT_VOTOS_NOMINAIS + QT_VOTOS_LEGENDA. For 2018 and 2022, the analysis uses QT_VOTOS_NOMINAIS_VALIDOS + QT_TOTAL_VOTOS_LEG_VALIDOS, which includes converted nominal-to-legend votes. Using the direct legend-vote column instead would exclude 206,687 votes in 2018 and 123,458 votes in 2022. Since the inversion criterion is a majority-threshold criterion, small vote-denominator errors can shift borderline cases across the 50 percent boundary. The audit decisions are reported in Appendix A.

Cabinet periods are linked to elections by mandate-window overlap: the 2014 election to cabinet periods from 2015 through 2018, the 2018 election to periods from 2019 through 2022, and the 2022 election to the available 2023–2026 window. Party labels are translated into election-year labels before coalition vote and seat totals are computed. This step is necessary because the analysis evaluates party sets as represented in the election that produced the Chamber delegation, not as they may have been reorganized afterward through mergers or renamings.

The ideology order uses party classifications from 2023 for the 2014 and 2018 elections and from 2025 for the 2022 election. This is a practical scope condition: retroactive ideology classifications are used to make the interval search tractable, which means the 2014 and 2018 interval results project later positional information onto earlier party labels. The cabinet analysis is not exposed to this concern, since it does not require ordering every party by ideology.

4 Party-level disproportionality

The coalition analysis begins with party-level seat differentials, which are the arithmetic substrate from which coalition inversions are assembled. Table 1 reports the largest positive and negative differentials by election. The table should not be read as a causal explanation of the inversions. It shows which party-level distortions were available for coalitional aggregation.

Table 1: Largest party-level seat differentials by election

Election	Direction	Party	Vote share	Seats	Quota	Seat diff.
2014	over	PMDB	11.09	65	56.87	8.13
2014	over	PSD	6.13	36	31.45	4.55
2014	over	PTB	4.02	25	20.63	4.37
2014	over	PR	5.79	34	29.70	4.30
2014	over	PP	6.60	38	33.88	4.12
2014	under	PSDB	11.39	54	58.43	-4.43
2014	under	PSOL	1.79	5	9.20	-4.20
2014	under	PT DO B	0.83	1	4.28	-3.28
2014	under	PSL	0.83	1	4.26	-3.26
2014	under	PV	2.06	8	10.56	-2.56
2018	over	PP	5.50	37	28.24	8.76
2018	over	PR	5.32	33	27.28	5.72
2018	over	MDB	5.54	34	28.40	5.60
2018	over	DEM	4.66	29	23.92	5.08
2018	over	PSD	5.85	35	30.01	4.99
2018	under	PSL	11.64	52	59.70	-7.70
2018	under	NOVO	2.80	8	14.35	-6.35
2018	under	PSOL	2.83	10	14.53	-4.53
2018	under	PV	1.62	4	8.32	-4.32
2018	under	PRTB	0.70	0	3.58	-3.58
2022	over	PL	16.61	99	85.22	13.78
2022	over	UNIÃO	9.34	59	47.90	11.10
2022	over	PT	12.10	69	62.06	6.94
2022	over	PP	7.95	47	40.76	6.24
2022	over	MDB	7.27	42	37.28	4.72
2022	under	PSOL	3.52	12	18.06	-6.06
2022	under	PTB	1.30	1	6.67	-5.67
2022	under	PSB	3.82	14	19.57	-5.57
2022	under	PODE	3.30	12	16.94	-4.94
2022	under	SOLIDARIEDADE	1.56	4	7.98	-3.98

Notes: Positive values indicate that the party received more seats than its national vote quota. Negative values indicate underrepresentation. The table reports the five largest positive and five largest negative party-level seat differentials in each election. Source artifact:

\ArtifactRoot/tables/table_02_party_over_underrepresentation.csv.

The pattern is not uniform across elections, and this non-uniformity matters for the inversion results. In 2014, the largest positive differentials are concentrated in PMDB, PSD, PTB, PR, and PP, precisely the parties that later enter or surround the Temer-period coalition space. In 2018, several parties associated with the Bolsonaro congressional support base are overrepresented, but as will be shown, the observed cabinet coalitions do not accumulate enough seats to cross the 257-seat threshold. In 2022, the largest positive differentials are concentrated in PL and UNIÃO, followed by PT, PP, and MDB. This is why the 2022 right-side ideological interval is especially

consequential: it combines parties with large positive differentials into a party set that crosses the seat-majority threshold without crossing the vote-majority threshold.

Figure 1 displays the underlying scatter. Points above the proportionality line represent parties that received more seats than their vote share would imply; points below represent parties that received fewer. The figure reveals that the seat differentials underlying the coalition inversions are not exceptional deviations. They are generated by the ordinary dispersion of parties around the proportionality line. The coalition-level inversion emerges not from a single anomalous party but from the accumulation of several moderate positive differentials.

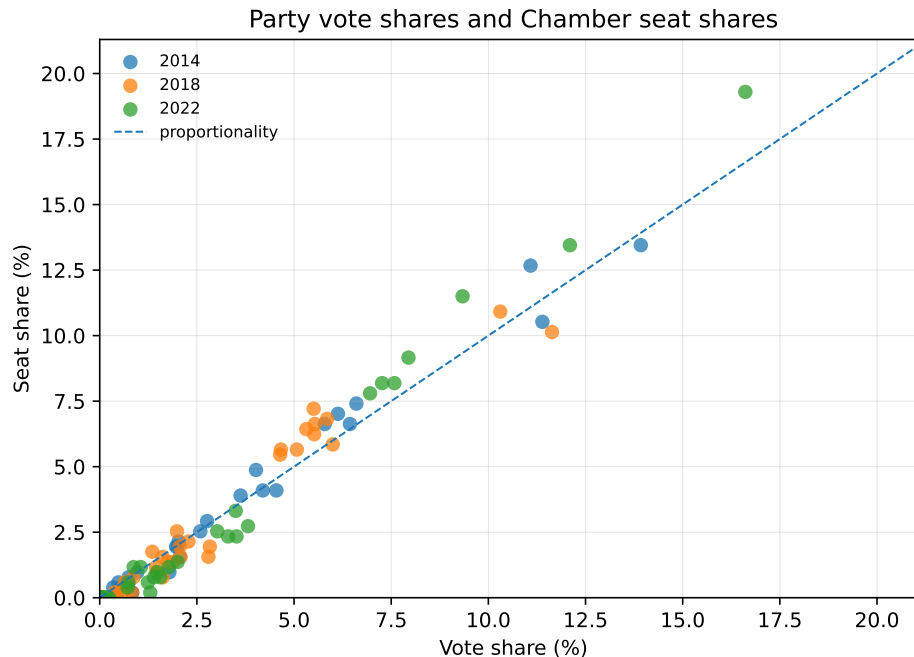


Figure 1: Party vote shares and Chamber seat shares

Notes: The dashed line is exact proportionality. Each point is a party in one election year. The vertical distance from the line is proportional to the party's seat differential; in seat units, it equals 513 times the vertical gap measured as a share. Source artifact: `\ArtifactRoot\figure_data\party_vote_share_vs_seat_share.csv`.

5 Observed cabinet coalitions

Observed cabinet coalitions are the central empirical objects because they are actual, institutionally relevant party sets. Many possible coalitions can be constructed from any fragmented election; most are politically meaningless. A cabinet-period coalition is different: it identifies a party set that organized or supported a specific government during a bounded historical period. The inversion question applied to such objects therefore has a concrete political referent.

Table 2 reports the three observed inversion cases. To clarify the classification: a coalition is *votes+seats* when it has both a vote majority and a seat majority; *votes only* when it has a vote majority without a seat majority; *seats only* when it has a seat majority without a vote

majority. The last category is the inversion category, and the three cases in the table are the only observed-coalition instances of it across the three elections and 22 cabinet periods covered.

Table 2: Observed cabinet-period coalition inversions

Election	Period	Days	Vote share	Seats	Seat diff.	Parties
2014	2016.2	2	46.54	259	20.27	PCdoB, PDT, PMDB, PR, PSD, PT, PTB
2014	2017.1	100	49.91	266	9.95	DEM, PMDB, PP, PPS, PSB, PSD, PSDB, PV
2022	2023.1	255	48.89	263	12.20	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO

Notes: Vote share is the coalition’s share of the national federal-deputy vote in the relevant election. Seat diff. is $s_C - q_C$. Source artifact: \ArtifactRoot/tables/table_04_observed_cabinet_inversions_only.csv.

The three cases differ in their political interpretation, and this heterogeneity is itself analytically relevant. The 2016.2 period lasted two days, spanning April 14–15, 2016, and is best treated as an event-window diagnostic rather than a stable governing configuration. It reveals that the coalition structure crossed the inversion threshold during the impeachment rupture, but it would be an overreach to interpret a 48-hour window as evidence of a systematic legislative majority. The 2017.1 result is the substantively important case from the 2014 election. The coalition of DEM, PMDB, PP, PPS, PSB, PSD, PSDB, and PV commanded 266 Chamber seats while corresponding to 49.91 percent of the national federal-deputy vote. Under the criterion developed above, this is a coalition inversion: the party set organizing the Temer government was institutionally amplified beyond its national vote share.

The 2022 result establishes that the inversion phenomenon is not reducible to the Temer episode or to the specific political crisis of 2016. The initial Lula coalition; MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, and UNIÃO; held 263 seats from January 1 to September 12, 2023, with 48.89 percent of the vote. This coalition subsequently expanded to include PP and Republicanos, shifting to the 2023.2 period with 63.79 percent of the vote and 350 seats. At that point the coalition had both a vote majority and a seat majority, so the inversion no longer applied. The transition illustrates a general pattern: the inversion is a transient property that depends on which parties are in the coalition at any moment, not a permanent feature of the chamber.

The negative result from the 2018 election is as important analytically as the positive cases. No observed cabinet coalition after the 2018 election satisfies the inversion criterion. The closest case is the 2021.3 coalition with 45.50 percent of the vote and 250 seats; a positive seat differential of 16.60 but still 7 seats short of the 257-seat threshold. The Bolsonaro-period coalitions are therefore overrepresented without being inversions, which confirms that overrepresentation and inversion are distinct conditions. A coalition can gain from the seat allocation procedure, in the sense of receiving more seats than its vote share implies, while remaining below the legislative majority threshold.

Figure 2 makes the temporal and cross-mandate structure visible. The two inversion periods from the 2014 mandate; the short 2016.2 window and the 100-day 2017.1 period; appear as moments in which the seat-share panel is above its threshold while the vote-share panel is below its threshold. The 2018-mandate panels show no such crossing, confirming the absence of observed inversions. The 2022-mandate panels show the 2023.1 inversion followed by the transition to the overrepresented but non-inversion 2023.2 coalition. A duration-weighted summary reinforces this reading: 102 of 1,461 covered days in the 2014 mandate are inversion days, zero of the covered

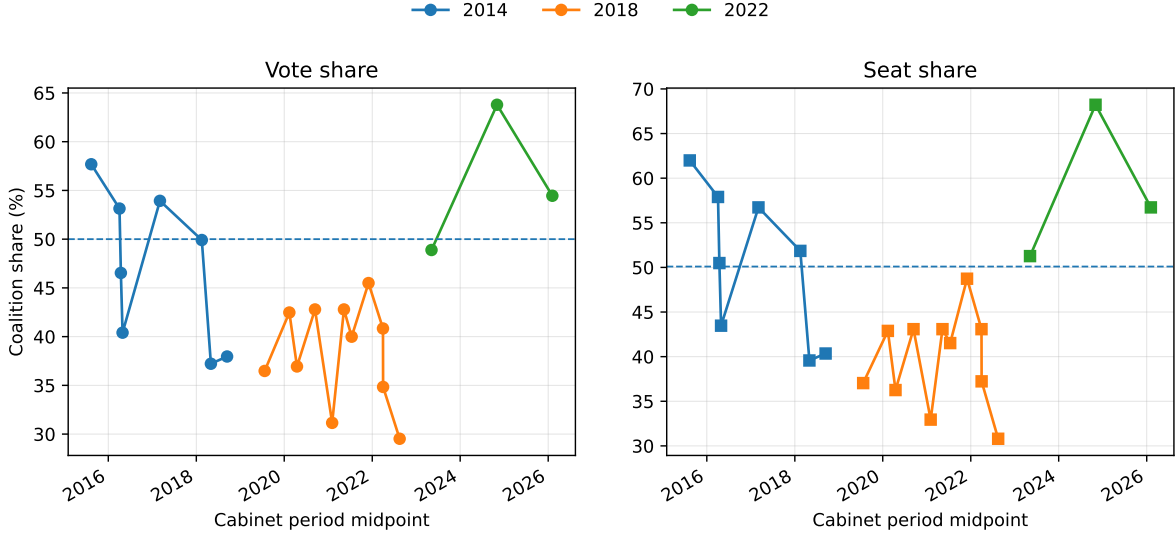


Figure 2: Observed cabinet-period coalition vote shares and seat shares

Notes: The left panel reports coalition vote shares and the right panel coalition seat shares over the mandate windows for each election year. The horizontal reference lines mark the vote-majority threshold of 50 percent and the seat-majority threshold of $257/513 \approx 50.10$ percent. Inversion periods are those in which the vote-share panel falls below its reference line while the seat-share panel is above its reference line. Source artifact:

\ArtifactRoot/figure_data/observed_coalition_timeline.csv.

days in the 2018 mandate, and 255 of 1,096 covered days in the available 2022 window.

6 Ideologically contiguous intervals

Observed cabinet coalitions identify politically realized inversions. A complementary question is whether the vote-seat geometry of the Chamber generates inversions for structured party sets that are not contingent on which governments actually formed. To answer this, I order parties by ideology and enumerate every contiguous no-gap interval. If the ordered parties are

$$p_1, p_2, \dots, p_n,$$

then an interval is any set

$$[p_i, p_j] = \{p_i, p_{i+1}, \dots, p_j\}, \quad i \leq j.$$

An interval is a coalition inversion when $s_{[p_i, p_j]} \geq 257$ and $v_{[p_i, p_j]}/V \leq .5$.

Definition 2 (Endpoint-minimal interval inversion). *An ideological interval $[p_i, p_j]$ is endpoint-minimal when it is a coalition inversion and removing either endpoint party destroys the seat majority. This is minimality under contiguous endpoint deletion, not minimality with respect to arbitrary deletion of interior parties.*

The endpoint-minimality criterion identifies the tightest cases. An endpoint-minimal interval is one where both boundary parties are necessary for the seat majority, which gives a more precise characterization of the coalition inversion than merely listing all inversion intervals regardless of

size. The definition is intentionally stricter than a generic search over all party subsets and less restrictive than a search over realized cabinets only.

Table 3: Ideological interval summary by election

Election	All intervals	Seat-majority intervals	Inversions	Minimal inversions
2014	528	180	8	4
2018	630	172	0	0
2022	528	99	6	2

Notes: Intervals are contiguous no-gap blocks in the party ideology order. Source artifact: `\ArtifactRoot\tables\table_05_ideological_interval_summary_by_election.csv`.

Table 3 shows the aggregate pattern. The 2018 result deserves particular attention. There are 172 seat-majority ideological intervals in 2018, yet none of them is an inversion. This means that every contiguous ideological bloc that controls a seat majority in 2018 also commands a vote majority. The absence of interval inversions in 2018 is therefore not attributable to the absence of large legislative blocs; it reflects a specific feature of the vote-seat relationship in that election. In 2014 and 2022, the seat-majority intervals are populated with inversions at their margins.

Table 4: Endpoint-minimal ideological interval inversions

Election	Start	End	Parties	Vote share	Seats	Seat diff.	Interval
2014	PSB	PTN	12	48.92	260	9.03	PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN
2014	PTB	PR	14	47.59	257	12.86	PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR
2014	PT DO B	PSDC	16	48.99	257	5.70	PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR, PRB, PTC, PSDC
2014	SOLIDARIEDADE	PSL	16	48.98	257	5.72	SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PPL, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL
2022	MDB	UNIÃO	15	49.98	265	8.59	MDB, PMN, PSDB, PSD, PMB, PODE, PROS, PRTB, AGIR, PTB, PP, DC, REPUBLICANOS, PSC, UNIÃO
2022	PP	PL	8	45.35	258	25.36	PP, DC, REPUBLICANOS, PSC, UNIÃO, PATRIOTA, NOVO, PL

Notes: The Parties column reports interval size. Vote share is the percentage of national federal-deputy votes. Source artifact: `\ArtifactRoot\tables\table_06_minimal_ideological_interval_inversions.csv`.

Table 4 reveals two distinct forms of the inversion. The 2014 endpoint-minimal intervals are close to the 50 percent vote threshold; ranging from 47.59 to 48.99 percent; and several are just at the 257-seat floor. These are genuine inversions, but they are proximity inversions in the sense expected by the theoretical literature on PR inversions: close cases that arise near the boundaries

of both thresholds (Miller 2015). The 2022 PP–PL interval is structurally different. It contains only eight parties, spans a compact right-side ideological range, and obtains a mere 45.35 percent of the national vote while holding 258 seats. A seat differential of 25.36 for an eight-party interval is a large concentrated distortion, not a boundary-proximity effect. This is the strongest structural finding in the analysis.

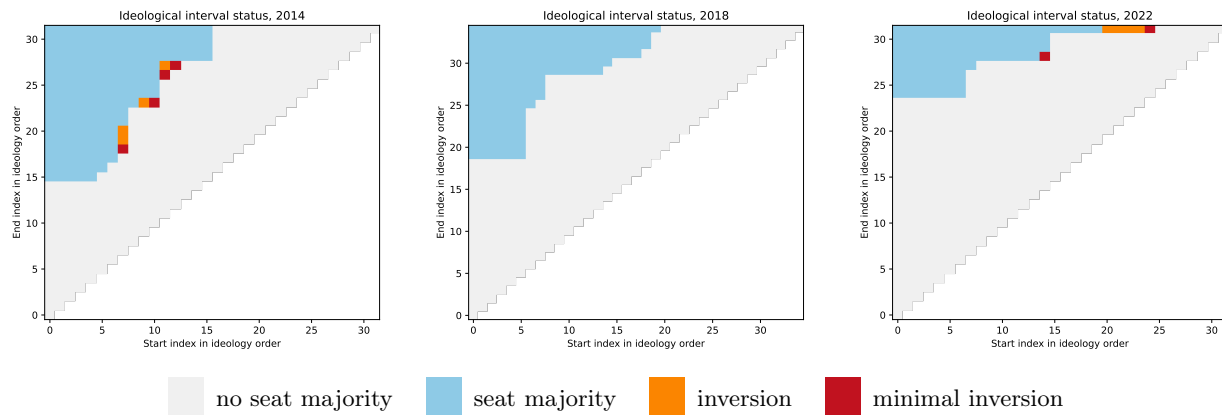


Figure 3: Ideological interval status by election

Notes: Each matrix enumerates contiguous ideological intervals by start and end index in the ideology order. Colors identify four discrete categories. The orange and red cells in 2014 and 2022 identify inversion intervals; their complete absence in the 2018 matrix confirms that no seat-majority ideological bloc in that election is an inversion. Source artifact: `\ArtifactRoot\figure_data\ideological_interval_heatmap.csv`.

Figure 3 makes the cross-election structure visible. The 2014 and 2022 matrices contain orange and red cells concentrated in specific ideological ranges; the center-right region in 2014, and both a broad center-right interval and a compact right-side interval in 2022. The 2018 matrix, despite having many seat-majority intervals (the blue cells), contains no inversion cells. The visual contrast between the 2018 matrix and the other two underscores that the inversion property is election-specific, not a mechanical consequence of chamber fragmentation.

7 Interpretation

The main finding is that Brazilian open-list proportional representation can translate nationally minority party sets into Chamber majorities. This result contributes to the literature on PR inversions by moving from formal possibility to empirical realization and structural characterization. Miller establishes that coalitionwise inversions can occur under highly proportional rules because the whole-number problem cannot be eliminated (Miller 2015). The Brazilian analysis shows that this possibility is realized in observed cabinet coalitions and embedded in the ideological geometry of the Chamber in two of the three elections examined.

The distinction between overrepresentation and inversion is worth emphasizing. A coalition with a positive seat differential is overrepresented: it receives more seats than its national vote share implies. But overrepresentation does not entail inversion; the differential must be large enough to cross the absolute majority threshold. The Bolsonaro-period coalitions from the 2018 election illustrate this. Several have positive seat differentials, and the closest case, the 2021.3 coalition, has a differential of 16.60 seats. Nevertheless, it holds only 250 seats; short of the 257-

seat floor. This confirms that the majority criterion performs genuine discriminatory work: it separates cases of ordinary proportionality deviation from cases in which the deviation changes the majority status of a party set.

The Temer-period finding warrants careful interpretation. The 2017.1 coalition controlled 266 seats with 49.91 percent of the national federal-deputy vote. Under the criterion developed here, this is a coalition inversion. What this supports is a narrow institutional claim: the party set organizing the Temer government was a Chamber majority without being a federal-deputy vote majority in the 2014 election-party space. What this does not support is any inference about whether voters would have endorsed or rejected that government in a direct plebiscite, whether the impeachment of Rousseff was democratically legitimate, or what the behavioral consequences of the majority configuration were for legislative discipline or policy outcomes. The analysis isolates the input-output relationship between vote aggregation and seat distribution. It does not model what happened inside the Chamber once the seat distribution was fixed.

The 2022 results complicate any narrowly partisan reading of the inversion phenomenon. The initial Lula coalition is itself an inversion, and the PP–PL interval is the strongest structural finding in the analysis. The phenomenon is therefore not specific to any president, ideological side, or political crisis. It is a structural possibility of the Brazilian Chamber under conditions of fragmented open-list PR and coalition formation, and it can be realized by party sets of very different political orientations.

The ideological interval analysis adds a structural layer to the cabinet results. Cabinet coalitions are not arbitrary, but neither do they exhaust the representational structure of the Chamber. A cabinet coalition may or may not coincide with a contiguous ideological interval, and an interval may reveal a legislative configuration that is structurally available without ever being realized as a governing arrangement. The two analyses are therefore answering different but complementary questions: the cabinet analysis identifies which politically realized party sets are inversions, while the interval analysis identifies which structured ideological blocks in the Chamber’s party space have the inversion property.

8 Limitations and extensions

Several limitations define the scope of what this analysis can and cannot establish. The first concerns the level of aggregation. The analysis uses national aggregate federal-deputy votes and national aggregate seats, which is appropriate for identifying coalition-level representation in the Chamber but does not decompose the inversion by state or district magnitude. A natural extension is to calculate each inversion coalition’s state-level contribution to its national seat differential. That would show whether particular states or district magnitudes drive the inversion, or whether it is distributed across the full federal structure.

The second limitation is the absence of a national PR counterfactual. A useful robustness exercise would reallocate all 513 seats as if Brazil were a single national district under D’Hondt, Sainte-Laguë, and Largest Remainder-Hare. This would separate the contribution of federalized district structure from pure national rounding and fragmentation effects. If an inversion disappears under national reallocation, then the district federalization is doing substantive representational work. If it persists, then the inversion is closer to the pure coalitionwise phenomenon that Miller identifies as inherent to any finite legislature.

The third limitation concerns the ideology order. The 2014 and 2018 orderings use a later classification source, projecting 2023 party positions onto earlier party labels. This is a necessary practical choice for making the interval search tractable, but it introduces a scope condition: the

interval results depend on the retroactive classification. The cabinet results are not exposed to this concern. Alternative ideology sources and classification approaches should be examined in subsequent work to assess the sensitivity of the interval findings.

The fourth limitation is that cabinet coalition membership is treated as a party-set object. The analysis does not model legislative discipline, individual deputy behavior, roll-call voting, or portfolio bargaining. A cabinet party set with 266 elected deputies is not equivalent to a guaranteed bloc of 266 disciplined votes. The criterion is representational; it asks about the seat share of the party set; not behavioral. This scope condition is shared with most coalition analysis in comparative politics, but it should be explicit because the object is the institutional translation of votes into seats, not the subsequent translation of seats into legislative outcomes.

The fifth limitation concerns the vote-coalition correspondence. The vote share attributed to a coalition is the sum of the party votes of its members, not a direct vote for the coalition as a collective object. A voter who supports a party that enters a cabinet coalition need not have supported, or even anticipated, that coalition. This is a structural feature of party-list systems and is not specific to the Brazilian case, but it limits what can be inferred about voter preferences over coalition governments. The analysis establishes that the party set was a vote minority and a seat majority; it does not establish that the coalition was rejected or endorsed by a majority of voters as a cabinet.

9 Conclusion

The question motivating this analysis was whether Brazilian proportional representation can generate coalition-level seat majorities without vote majorities. The answer is yes, and the phenomenon is not confined to a single election or a single political emergency.

Three observed cabinet periods satisfy the inversion criterion. The 2017.1 coalition, which organized the Temer government, held 266 seats with 49.91 percent of the national federal-deputy vote for 100 days. The 2023.1 coalition, which organized the initial Lula government after the 2022 election, held 263 seats with 48.89 percent of the vote for 255 days. Conversely, no Bolsonaro-period coalition after the 2018 election reaches the seat-majority threshold, despite overrepresentation, confirming that the majority criterion does genuine discriminatory work. The ideological interval results confirm the pattern and identify its structural extent. In 2014 and 2022, contiguous ideological blocks can become seat majorities without vote majorities. In 2018, no such block exists. The 2022 PP–PL interval; eight parties, 45.35 percent of the vote, 258 seats; is difficult to dismiss as a boundary-proximity artifact.

The broader implication is that proportionality should not be evaluated only at the party level. In fragmented systems, the politically operative unit is often a coalition, and the accumulation of party-level seat differentials can change the majority status of that coalition. Brazilian open-list PR does not merely translate votes into party seats. It supplies the institutional raw material from which governments and ideological blocs assemble legislative majorities. Under some configurations, that assembly creates seat majorities without vote majorities; a structural feature of the institution, not an anomaly of any particular political moment.

The next analytical step is to locate the mechanism more precisely. State-level decomposition and national PR counterfactuals can determine how much of each inversion comes from federalized district allocation, how much from ordinary rounding and fragmentation, and whether some states contribute disproportionately to the distortion. Once that is done, the analysis can connect the institutional fact established here; that nationally minority party sets can become Chamber majorities; to a sharper account of which features of Brazil’s electoral architecture produce the

effect.

A Artifact inventory and audit decisions

The main files used in the analysis are:

- Party vote and seat totals by election, from TSE filings.
- Observed cabinet-period party composition and duration data.
- Cabinet-to-election party label crosswalk.
- Ideological interval enumeration by election.
- Vote-column audit crosswalk, summarized in Table 5.

Table 5: Vote-column audit rules

Election	Nominal vote column	Legend vote column	Difference from legacy
2014	QT_VOTOS_NOMINAIS	QT_VOTOS_LEGENDA	n/a
2018	QT_VOTOS_NOMINAIS_VALIDOS	QT_TOTAL_VOTOS_LEG_VALIDOS	206,687
2022	QT_VOTOS_NOMINAIS_VALIDOS	QT_TOTAL_VOTOS_LEG_VALIDOS	123,458

Notes: The difference column reports additional votes included in 2018 and 2022 relative to using the direct valid legend-vote column. The larger column is used throughout to minimize denominator error near the 50 percent inversion boundary. Source artifact: \ArtifactRoot/tables/table_07_audit_vote_columns_crosswalk.csv.

B Full observed cabinet-period table

Table 6: Observed cabinet-period coalition summary

Election	Period	Days	Vote share	Seats	Seat diff.	Status
2014	2015.1	448	57.69	318	22.07	votes+seats
2014	2016.1	21	53.14	297	24.39	votes+seats
2014	2016.2	2	46.54	259	20.27	seats only
2014	2016.3	26	40.41	223	15.72	neither
2014	2016.4	595	53.93	291	14.33	votes+seats
2014	2017.1	100	49.91	266	9.95	seats only
2014	2018.1	51	37.22	203	12.07	neither
2014	2018.2	218	37.96	207	12.26	neither
2018	2019.1	406	36.48	190	2.86	neither
2018	2020.1	4	42.47	220	2.11	neither
2018	2020.2	123	36.94	186	-3.50	neither
2018	2020.3	176	42.79	221	1.49	neither
2018	2020.4	109	31.15	169	9.19	neither
2018	2021.1	87	42.79	221	1.49	neither

Election	Period	Days	Vote share	Seats	Seat diff.	Status
2018	2021.2	41	39.99	213	7.84	neither
2018	2021.3	239	45.50	250	16.60	neither
2018	2022.1	1	40.83	221	11.52	neither
2018	2022.2	1	34.84	191	12.28	neither
2018	2022.3	274	29.52	158	6.55	neither
2022	2023.1	255	48.89	263	12.20	seats only
2022	2023.2	833	63.79	350	22.76	votes+seats
2022	2025.1	86	54.45	291	11.66	votes+seats

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